Sean Luka Girgis

Senior Python / SQL Developer | Financial Data, Pandas, Unix & Data Integration

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Python and SQL developer with 20+ years of enterprise technology experience, including regulated financial-services data workflows, Python/Pandas, NumPy, SQL, Unix/Linux, data ingestion, data integration, troubleshooting, production support, and release-oriented delivery. Strong background building data processing pipelines, validation workflows, reporting systems, and operational automation for large-scale enterprise environments. Best aligned to Python data/infrastructure roles requiring Pandas/NumPy, SQL, Unix, REST API concepts, object-oriented design, data structures, data ingestion, system stability, and independent problem solving; FastAPI, async/await, and quantitative-investing infrastructure are positioned as adjacent or active-learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark data pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational feeds into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Developed Python and SQL processing workflows for data ingestion, cleansing, enrichment, reconciliation, validation, and delivery to downstream reporting and analytics consumers in a regulated financial-services environment.
- Used Pandas, Python data structures, SQL, and reusable processing patterns to normalize large operational datasets and prepare model-ready inputs for Prophet and scikit-learn forecasting workflows.
- Designed AWS and database-backed data platform components using S3, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, and analytics workloads.
- Built data quality and reconciliation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and trust in data outputs.
- Troubleshot production data and reporting issues by tracing ingestion feeds, transformation logic, SQL behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Documented transformation rules, validation checkpoints, metric definitions, operational runbooks, support procedures, and release notes to improve repeatability and system support.
- Collaborated with infrastructure, application, analytics, and business teams to deliver stable, secure, and maintainable data workflows aligned with enterprise technology standards.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.

- Supported production issue analysis by correlating monitoring signals, transaction behavior, and system performance data.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, dashboards, operational analytics, and production reliability.
- Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed reusable scripts, dashboards, alert logic, SLA thresholds, technical documentation, and operational standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze telemetry, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java

Flagship Projects

AI-Powered Job Search Pipeline
Built a Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact

generation, quality gates, document rendering, and application tracking.
Emphasized structured data processing, validation, API integration, and reliable workflow orchestration.
Technologies: Python, JSON, REST/API concepts, FAISS, Sentence Transformers, Grok / xAI API, Quality Gates

HorizonScale — Capacity Forecasting Automation
Built Python/Pandas/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights through repeatable reporting and dashboard workflows.
Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Streamlit

Enterprise APM Reporting Automation
Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance telemetry across large CA APM enterprise monitoring environments.
Technologies: KornShell/ksh, Perl, Unix/Linux, CA APM, SQL, Automation, Monitoring